



Presented by Aryan Pattnaik



Atliq Hardwares

# CONSUMER GOODS AD-HOC INSIGHTS

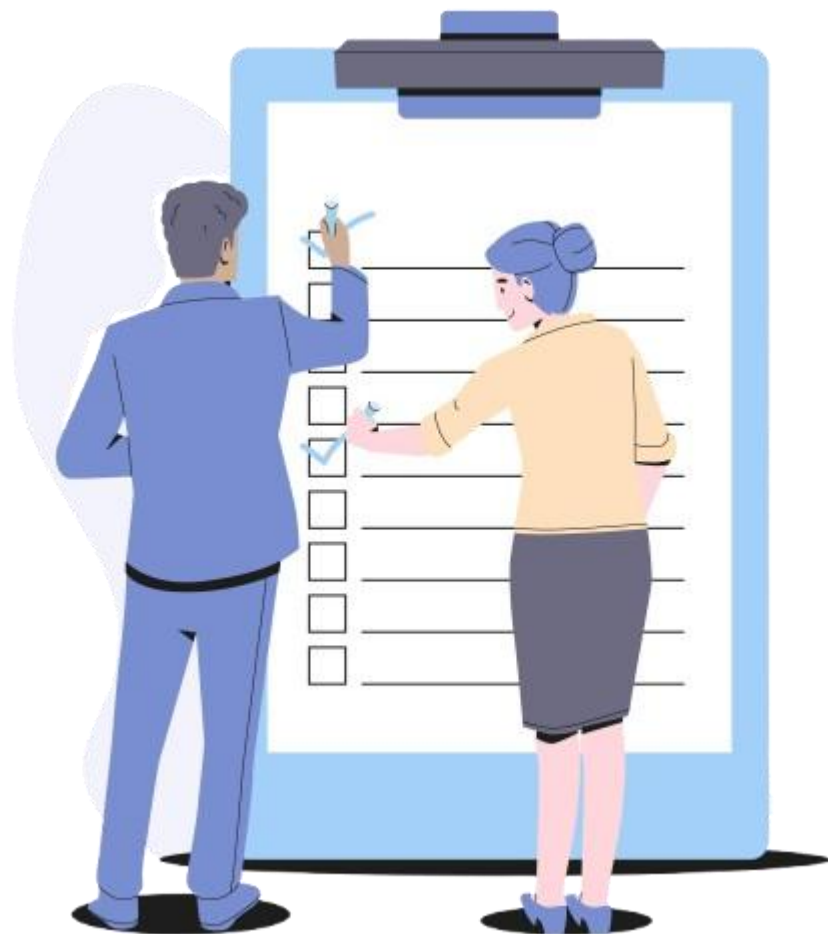
SQL Project Challenge

# AGENDA

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1. Background/Context
2. Getting familiar with Atliq's Business - Their Markets and Product lines
3. Getting familiar with the input data
4. Ad-hoc requests along with the queried result, visualizations and insights





# BACKGROUND/CONTEXT

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**Our Company** - Atliq Hardwares (imaginary company) - One of the leading computer hardware producers in India.

**Background-** The management noticed that they do not get enough insights to make quick and smart data-informed decisions.

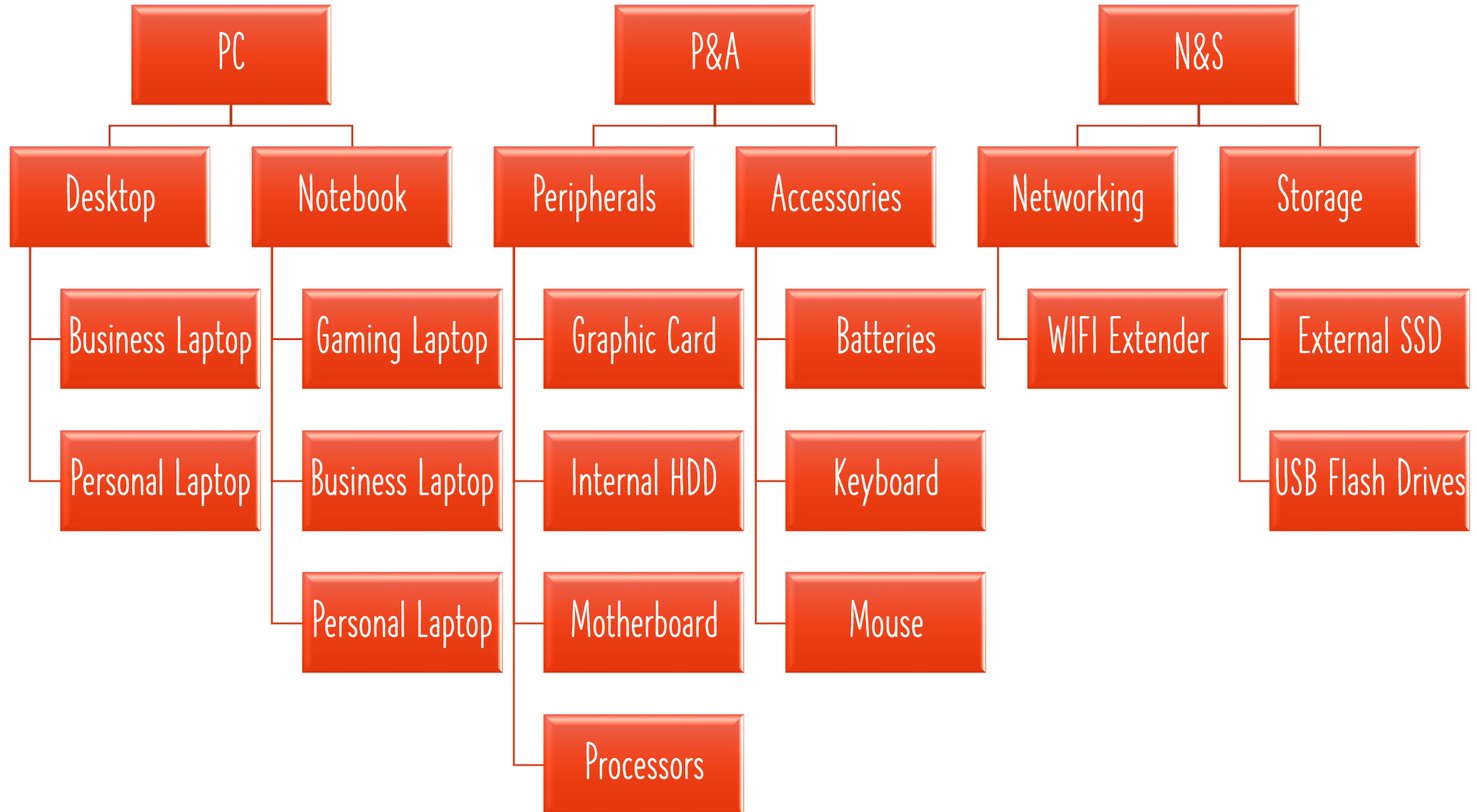
**Problem-** There are 10 ad-hoc requests for which the company needs insights.

**Approach** - Run a SQL query to answer these requests. Convert it into visualizations and present the Insights to the toplevel management.

# ATLIQ'S MARKETS



# PRODUCT LINES





Input data consists of sales data for FY 2020 and FY 2021, along with different other dimension tables like customer details, product details, etc.



| dim_customer          |
|-----------------------|
| customer_code INT     |
| customer VARCHAR(150) |
| platform VARCHAR(45)  |
| channel VARCHAR(45)   |
| market VARCHAR(45)    |
| sub_zone VARCHAR(45)  |
| region VARCHAR(45)    |

| fact_pre_invoice_deductions           |
|---------------------------------------|
| customer_code INT                     |
| fiscal_year YEAR                      |
| pre_invoice_discount_pct DECIMAL(5,4) |

| fact_gross_price          |
|---------------------------|
| product_code VARCHAR(45)  |
| fiscal_year YEAR          |
| gross_price DECIMAL(15,4) |

| dim_product              |
|--------------------------|
| product_code VARCHAR(45) |
| division VARCHAR(45)     |
| segment VARCHAR(45)      |
| category VARCHAR(45)     |
| product VARCHAR(200)     |
| variant VARCHAR(45)      |

| fact_sales_monthly       |
|--------------------------|
| date DATE                |
| product_code VARCHAR(45) |
| customer_code INT        |
| sold_quantity INT        |
| fiscal_year YEAR         |

| fact_manufacturing_cost          |
|----------------------------------|
| product_code VARCHAR(45)         |
| cost_year YEAR                   |
| manufacturing_cost DECIMAL(15,4) |

Ad-hoc requests  
Queried Results  
Insights  
And  
Visualization



## Request 1 :-

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

## Output :-

|   | market      |
|---|-------------|
| ▶ | India       |
|   | Indonesia   |
|   | Japan       |
|   | Philippines |
|   | South Korea |
|   | Australia   |
|   | Newzealand  |
|   | Bangladesh  |



## Insights :-

This analysis identifies all APAC markets where the customer *Atliq Exclusive* operates. It helps understand geographic presence and regional coverage of the brand within the APAC business unit. Our Exclusive store has established its presence in 8 major markets.



## Request 2 :-

What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields

- unique\_products\_2020
- unique\_products\_2021
- percentage\_chg

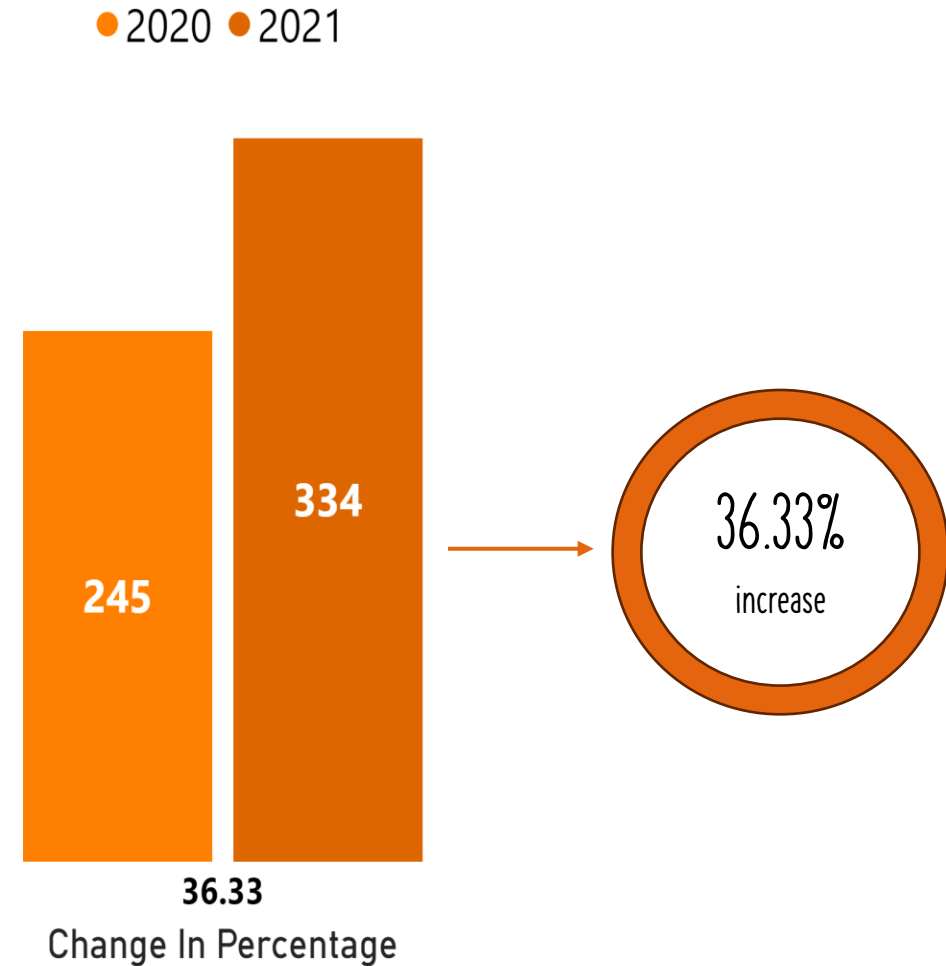
## Output :-

|   | unique_products_2020 | unique_products_2021 | percentage_chg |
|---|----------------------|----------------------|----------------|
| ▶ | 245                  | 334                  | 36.33          |

# Insights :-

This analysis measures product portfolio expansion by comparing the number of unique products launched in 2021 versus 2020. The percentage change indicates growth in product diversification strategy.

It's a good sign that we are continuously innovating and introducing new products to the market. In FY 2020, we had a total of 245 products, but in FY 2021, our count increased by 36% to 334 products.



## Request 3 :-

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

The final output contains 2 fields

- segment
- product\_count

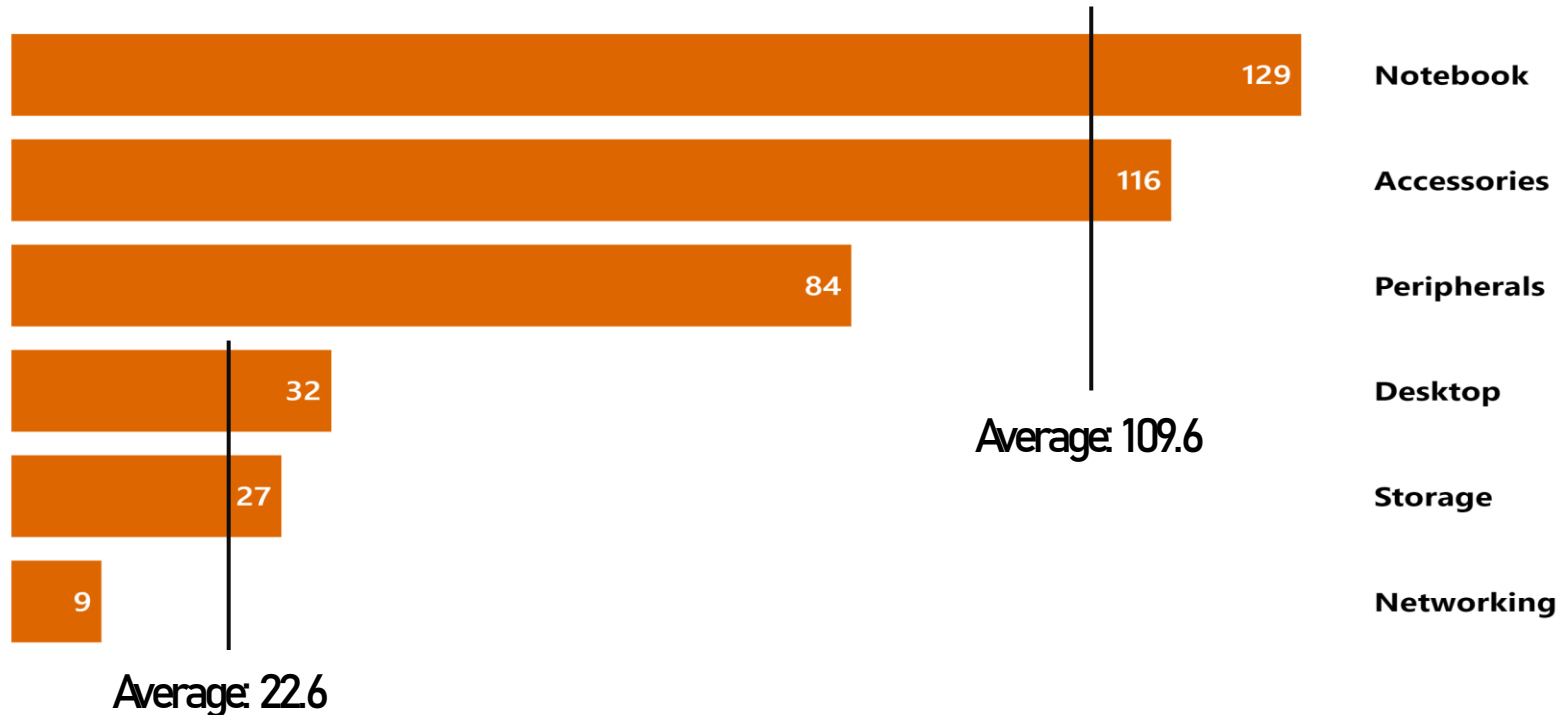
## Output :-

|   | segment     | product_count |
|---|-------------|---------------|
| ▶ | Notebook    | 129           |
|   | Accessories | 116           |
|   | Peripherals | 84            |
|   | Desktop     | 32            |
|   | Storage     | 27            |
|   | Networking  | 9             |

# Insights :-

This report shows the distribution of products across different segments. It highlights which segments contribute most to the overall product portfolio.

We provide a wide range of products under the segments Notebook, Peripherals, and Accessories, with an average of 110 products in each segment. However we still need to diversify our production in the Desktop, Networking, and Storage segments, where there are just an average of only 23 products per segment.



## Request 4 :-

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

The final output contains these fields

- segment product\_count\_2020
- product\_count\_2021
- difference

## Output :-

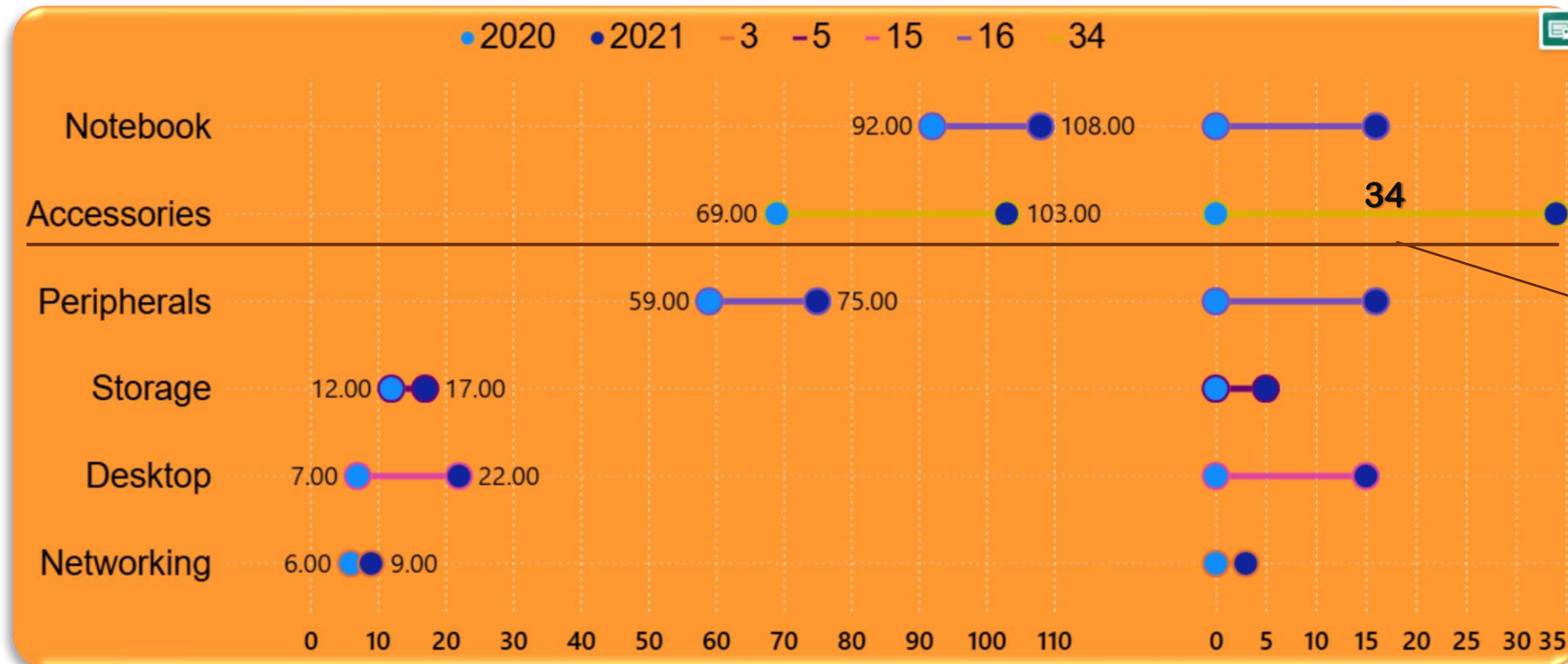
|   | segment     | product_count_2020 | product_count_2021 | difference |
|---|-------------|--------------------|--------------------|------------|
| ► | Accessories | 69                 | 103                | 34         |
|   | Notebook    | 92                 | 108                | 16         |
|   | Peripherals | 59                 | 75                 | 16         |
|   | Desktop     | 7                  | 22                 | 15         |
|   | Storage     | 12                 | 17                 | 5          |
|   | Networking  | 6                  | 9                  | 3          |



# Insights :-

This comparison identifies which product segment experienced the highest increase in unique products between 2020 and 2021, indicating strategic expansion focus areas.

In 2021, we were mainly focusing on diversifying our accessories segment. We introduced 34 new products to the market in accessories.



This segment had the most increase

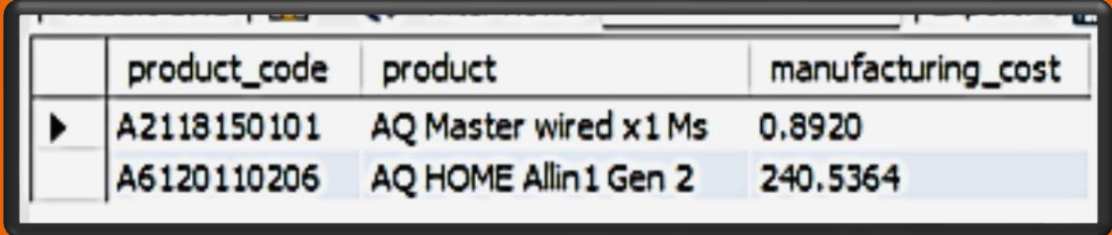
## Request 5 :-

Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields

- product\_code
- product
- manufacturing\_cost

## Output :-

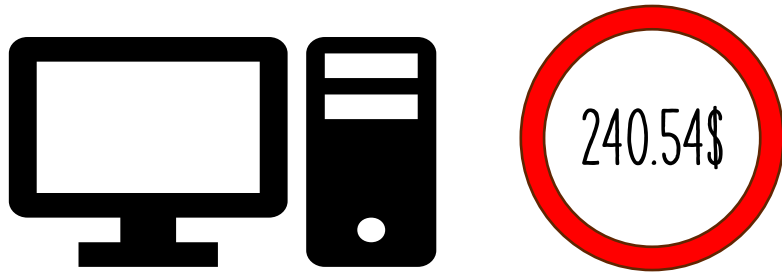


|   | product_code | product               | manufacturing_cost |
|---|--------------|-----------------------|--------------------|
| ▶ | A2118150101  | AQ Master wired x1 Ms | 0.8920             |
|   | A6120110206  | AQ HOME Allin1 Gen 2  | 240.5364           |

# Insights :-

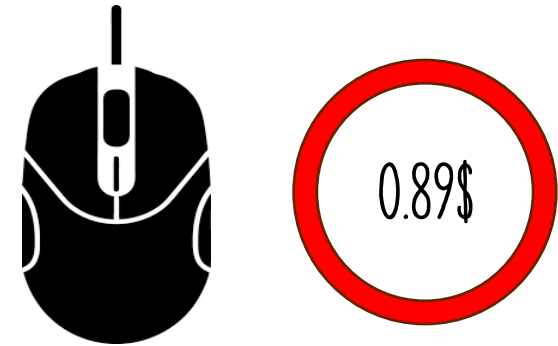
This analysis identifies products with the highest and lowest manufacturing costs. It helps evaluate cost efficiency and potential margin impact across the product portfolio.

## The Highest Manufacturing Cost



**Personal Desktop**  
AQ HOME Allin1 Gen 2 (Plus 3)

## The Lowest Manufacturing Cost



**Mouse**  
AQ Master wired x1 Ms (Standard 1)

## Request 6 :-

Generate a report which contains the top 5 customers who received an average high pre\_invoice\_discount\_pct for the fiscal year 2021 and in the Indian market.

The final output contains these fields

- customer\_code
- customer
- average\_discount\_percentage

## Output :-

|   | customer_code | customer | average_discount_percentage |
|---|---------------|----------|-----------------------------|
| ▶ | 90002009      | Flipkart | 0.3083                      |
|   | 90002006      | Viveks   | 0.3038                      |
|   | 90002003      | Ezone    | 0.3028                      |
|   | 90002002      | Croma    | 0.3025                      |
|   | 90002016      | Amazon   | 0.2933                      |

# Insights :-

This report highlights the top 5 customers in India who received the highest average pre-invoice discount in FY2021. It helps assess discount concentration and pricing strategy effectiveness.

In 2021, we offered nearly equal pre-invoice discount percentages to each of our top 5 customers, given that Flipkart is the most discounted customer in the Indian market, which equals 30.83%.



Amazon

29.33%

## Request 7 :-

Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns:

- Month
- Year
- Gross sales Amount

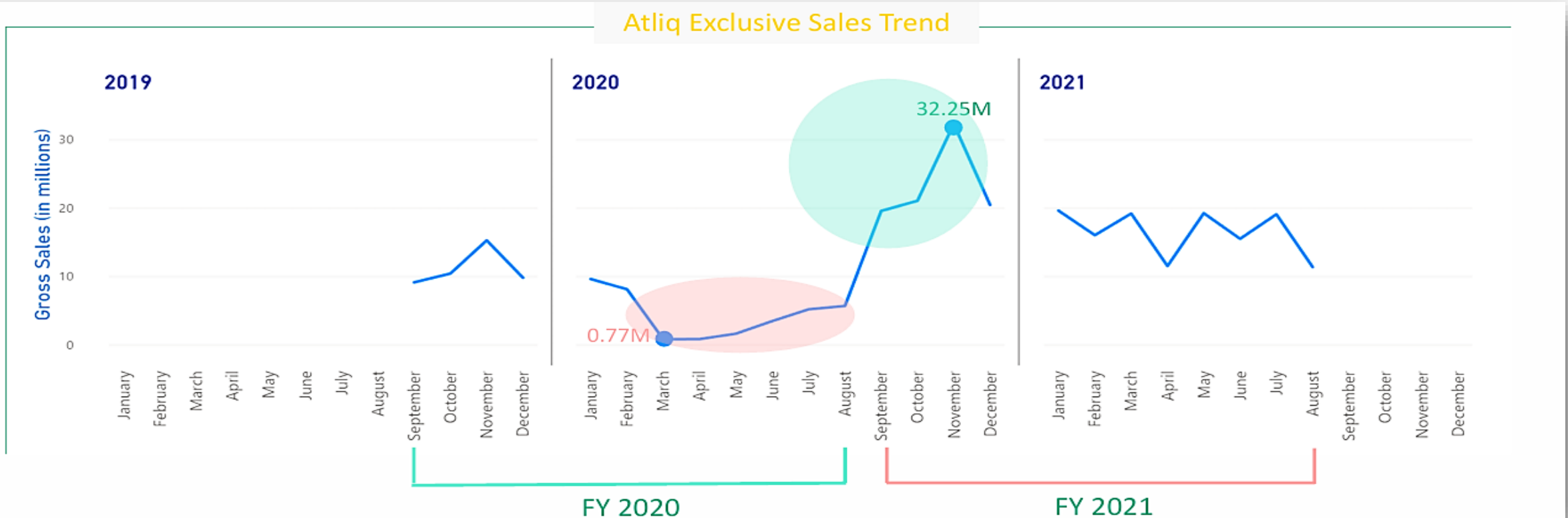
## Output :-

|   | Month     | Year | Gross_Sales_Amount |
|---|-----------|------|--------------------|
| ► | September | 2019 | 4496259.67         |
|   | October   | 2019 | 5135902.35         |
|   | November  | 2019 | 7522892.56         |
|   | December  | 2019 | 4830404.73         |
|   | January   | 2020 | 4740600.16         |
|   | February  | 2020 | 3996227.77         |
|   | March     | 2020 | 378770.97          |
|   | April     | 2020 | 395035.35          |
|   | May       | 2020 | 783813.42          |
|   | June      | 2020 | 1695216.60         |
|   | July      | 2020 | 2551159.16         |
|   | August    | 2020 | 2786648.26         |
|   | September | 2020 | 12353509.79        |
|   | October   | 2020 | 13218636.20        |
|   | November  | 2020 | 20464999.10        |
|   | December  | 2020 | 12944659.65        |
|   | January   | 2021 | 12399392.98        |
|   | February  | 2021 | 10129735.57        |
|   | March     | 2021 | 12144061.25        |
|   | April     | 2021 | 7311999.95         |
|   | May       | 2021 | 12150225.01        |
|   | June      | 2021 | 9824521.01         |
|   | July      | 2021 | 12092346.32        |
|   | August    | 2021 | 7178707.59         |

# Insights :-

This report analyze the monthly gross sales trend of Atliq Exclusive. It provides insights into seasonality, peak sales months, and performance fluctuations throughout the year.

For AtliqExclusive, November 2020 marked the highest sales, and March 2020 marked the lowest gross sales. It's very evident that the lower sales between March and August are because of COVID-19. However, it's a very good sign that the sales increased quickly after August and reached the highest level since the last two years in November.





## Request 8 :-

In which quarter of 2020, got the maximum total\_sold\_quantity?

The final output contains these fields sorted by the total\_sold\_quantity,

- Quarter
- total\_sold\_quantity

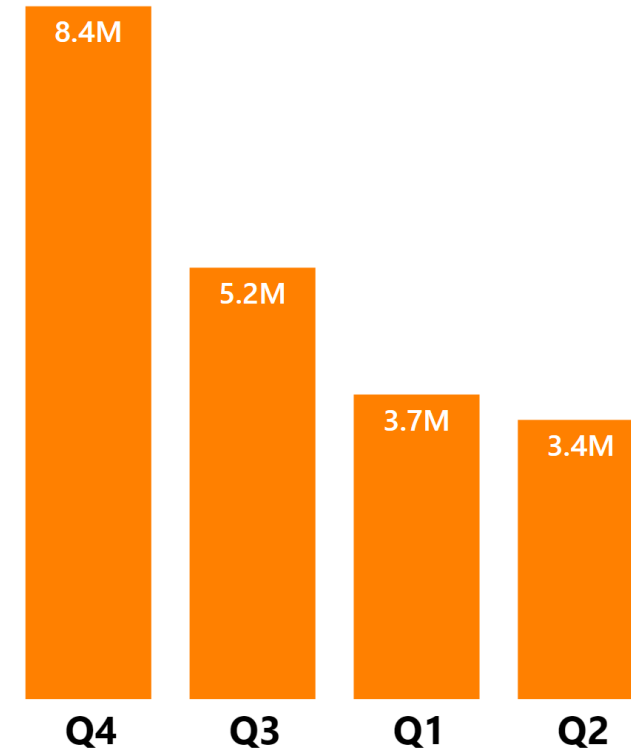
## Output :-

|   | Quarter | total_sold_quantity |
|---|---------|---------------------|
| ► | Q4      | 8425822             |
|   | Q3      | 5246770             |
|   | Q1      | 3704398             |
|   | Q2      | 3395899             |

## Insights :-

This analysis determines which quarter recorded the highest total sold quantity in 2020. It helps identify peak demand periods and seasonal sales patterns.

This again complements the previous insight. That is the effect of COVID-19 on our sales. The sold quantity decreased to 3.4 million in quarter 2 of FY 2020, when COVID-19 was at its peak. But we started recovering very early despite the continuance of the pandemic. This early recovery during quarter 4 is probably because of the increased need for hardware like desktops and notebooks as majority of the students began or continued to do their coursework online during this time, and there was a huge demand for computer accessories during this period.



## Request 9 :-

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

The final output contains these fields

- channel
- gross\_sales\_mln
- percentage

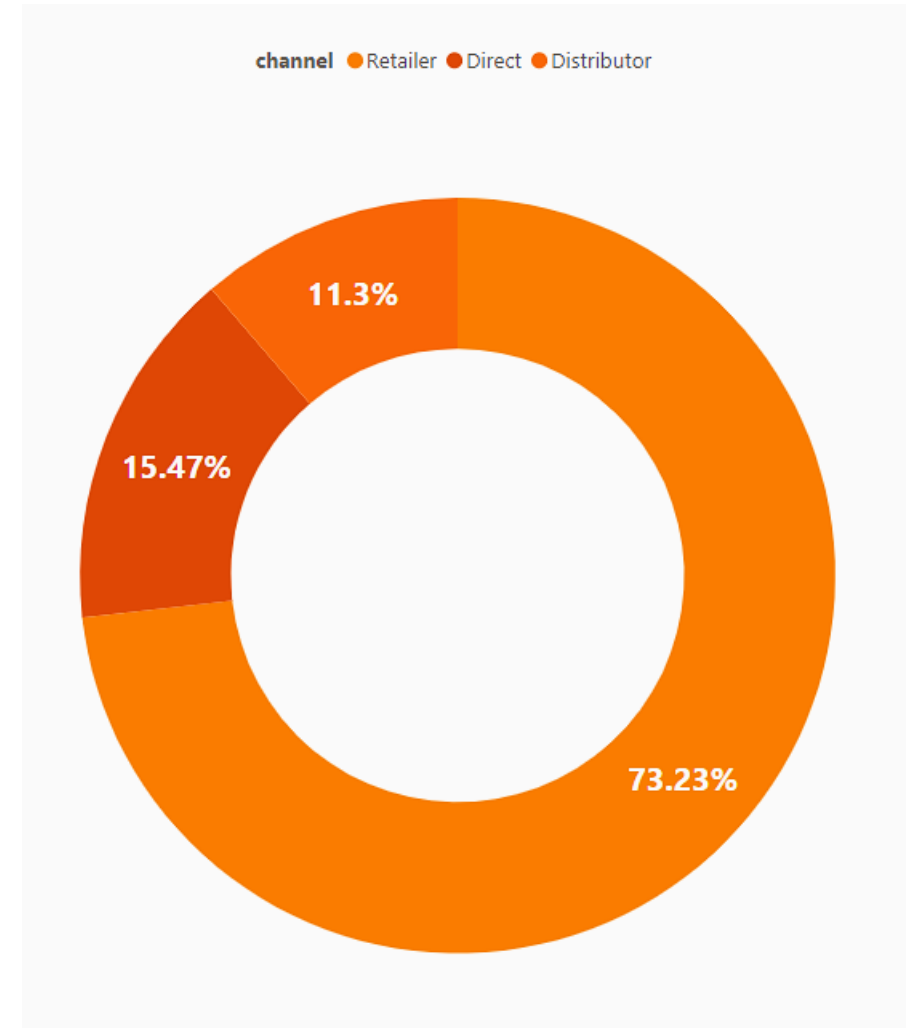
## Output :-

|   | channel     | gross_sales_mln | percentage |
|---|-------------|-----------------|------------|
| ▶ | Retailer    | 1219.08         | 73.23      |
|   | Direct      | 257.53          | 15.47      |
|   | Distributor | 188.03          | 11.30      |

# Insights :-

This report evaluates which sales channel contributed the highest revenue in FY2021 and calculates its percentage contribution to total sales.

The majority of our sales took place via retailers, which is 73% of the total sales. Only a very small percentage of our sales happened through direct and distributor channels.



## Request 10 :-

Get the Top 3 products in each division that have a high total\_sold\_quantity in the fiscal\_year 2021?

The final output contains these fields

- division
- product\_code
- product
- total\_sold\_quantity
- rank\_order

## Output :-

|   | division | product_code | product             | total_sold_quantity | rank_order |
|---|----------|--------------|---------------------|---------------------|------------|
| ▶ | N & S    | A6720160103  | AQ Pen Drive 2 IN 1 | 701373              | 1          |
|   | N & S    | A6818160202  | AQ Pen Drive DRC    | 688003              | 2          |
|   | N & S    | A6819160203  | AQ Pen Drive DRC    | 676245              | 3          |
|   | P & A    | A2319150302  | AQ Gamers Ms        | 428498              | 1          |
|   | P & A    | A2520150501  | AQ Maxima Ms        | 419865              | 2          |
|   | P & A    | A2520150504  | AQ Maxima Ms        | 419471              | 3          |
|   | PC       | A4218110202  | AQ Digit            | 17434               | 1          |
|   | PC       | A4319110306  | AQ Velocity         | 17280               | 2          |
|   | PC       | A4218110208  | AQ Digit            | 17275               | 3          |

# Insights :-

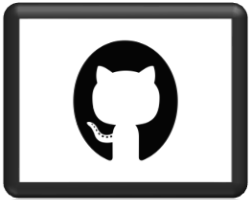
This analysis identifies the top three best-performing products in each division based on total sold quantity in FY2021. It highlights key revenue-driving products within each division.

- The top 3 selling products in N&S were **pen drives**, which were around **7 lakh in quantity**.
- The top 3 selling products in P&A were **mouse**, which were around **4 lakh in quantity**.
- The top 3 selling products in PC were **personal laptops**, which were around **17000 in quantity**.

## Top 3 Products by Division – FY 2021

| Division                   | Total Sold Quantity |
|----------------------------|---------------------|
| <b>N &amp; S</b>           | 2065621             |
| <b>AQ Pen Drive 2 IN 1</b> |                     |
| A6720160103                | 701373              |
| <b>AQ Pen Drive DRC</b>    |                     |
| A6818160202                | 688003              |
| A6819160203                | 676245              |
| <b>P &amp; A</b>           | 1267834             |
| <b>AQ Gamers Ms</b>        |                     |
| A2319150302                | 428498              |
| <b>AQ Maxima Ms</b>        |                     |
| A2520150501                | 419865              |
| A2520150504                | 419471              |
| <b>PC</b>                  | 51989               |
| <b>AQ Digit</b>            |                     |
| A4218110202                | 17434               |
| A4218110208                | 17275               |
| <b>AQ Velocity</b>         |                     |
| A4319110306                | 17280               |
| <b>Total</b>               | 3385444             |

# THANK YOU



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Gmail



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